UNIT – V : INPUT / OUTPUT (I/O) ORGANIZATION

Input / Output: Peripheral devices, I/O interface, I/O ports, Interrupts: interrupt hardware, types of interrupts and exceptions. Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access., I/O channels and processors. Serial Communication: Synchronous & asynchronous communication, standard communication interfaces.

◇ 1. Introduction to I/O System

The **Input/Output (I/O) system** connects the **CPU** and **main memory** with the **outside world** — such as keyboard, mouse, monitor, printer, etc.

➤ Role of I/O system:

- Helps CPU communicate with external devices.
- Transfers data between memory and peripherals.
- Improves overall system performance.

Example:
When you type on the keyboard (input device), data goes to CPU → memory → output shown on monitor.

◇ 2. Peripheral Devices

Peripheral devices are the external hardware components connected to the computer.

➤ Types of Peripheral Devices:

Type	Example	Function
Input Devices	Keyboard, Mouse, Scanner	Send data to the computer
Output Devices	Monitor, Printer, Speaker	Display or produce results
Storage Devices	Hard disk, Pen drive, CD	Store data permanently

Note: Peripherals are not part of the main CPU or memory, but they help in I/O operations.

◇ 3. I/O Interface

Since **CPU** and **I/O devices** work differently (speed, data size, format), an **interface** is required between them.

➤ I/O Interface Functions:

1. Acts as a **bridge** between CPU and peripheral devices.
2. Converts signals between **digital (CPU)** and **device-specific format**.
3. Handles control, timing, and data transfer.

Example:

When CPU sends data to printer — the I/O interface converts digital data into printer-understandable form.

◇ 4. I/O Ports

An **I/O Port** is a connection point or address through which CPU communicates with external devices.

➤ Types:

- **Parallel Port** → Transfers multiple bits at a time.

Example: Used for printers (old systems).

- **Serial Port** → Transfers data bit by bit.

Example: Used in modern systems like USB, COM ports.

◇ 5. Interrupts

➤ Definition:

An **interrupt** is a signal that temporarily stops CPU's current execution and asks it to handle a specific event.

Example:

When you press a key, keyboard sends an interrupt to CPU → CPU stops what it was doing → reads the key value.

➤ Need for Interrupts:

- Avoids continuous checking of devices (saves CPU time).
- Allows CPU to perform other tasks until a device needs attention.

➤ Interrupt Hardware:

It includes:

1. **Interrupt Request Line (IRQ)** – carries the interrupt signal to CPU.
2. **Interrupt Vector Table** – stores the address of the service routine.

3. **Interrupt Service Routine (ISR)** – the code that runs when interrupt occurs.

➤ Types of Interrupts:

Type	Description	Example
Hardware Interrupt	Generated by hardware device	Keyboard input, printer ready
Software Interrupt	Generated by a program or instruction	System calls, traps
Internal Interrupt	Due to internal errors	Divide by zero
External Interrupt	From external devices	Mouse click, disk request

➤ Exceptions:

Exceptions are **unexpected events** that occur during program execution (like errors).

Type	Example
Fault	Page fault (memory not available)
Trap	Breakpoint in debugging
Abort	Hardware failure

◇ 6. Modes of Data Transfer

Different ways CPU and I/O devices transfer data:

Mode	Description	Example	Advantage
Programmed I/O	CPU controls all transfer operations	CPU reads keypress one by one	Simple to implement
Interrupt-initiated I/O	Device sends interrupt when ready	Printer sends interrupt after printing	CPU not busy waiting
Direct Memory Access (DMA)	Data moves directly between memory and I/O without CPU	Disk → Memory transfer	Very fast, CPU free

Example to understand:

Suppose data needs to be read from disk → memory.

- **Programmed I/O:** CPU checks disk repeatedly → slow.
- **Interrupt I/O:** Disk interrupts CPU when data ready → better.
- **DMA:** Data transfers directly between disk and memory → best.

◇ 7. I/O Channels and Processors

➤ I/O Channel:

A special **hardware unit** that manages multiple I/O devices independently from CPU.

- Works like a **mini-CPU** for I/O operations.
- Reduces CPU load.

Example:

In mainframes, I/O channels handle disk and tape drives while CPU focuses on computation.

➤ I/O Processor (IOP):

- A separate processor for input/output tasks.
- Executes its own instructions for I/O operations.
- Communicates with main CPU via shared memory.

Think of it as:

CPU = main brain, IOP = assistant handling all input/output tasks.

◇ 8. Serial Communication

➤ Definition:

Data is sent **bit-by-bit** over a single line.

Used for **long-distance communication** (less wires, less cost).

➤ Types of Serial Communication:

Type	Description	Synchronization	Example
Synchronous	Data sent in a continuous stream with clock signal	Sender & receiver synchronized by clock	Ethernet, SPI
Asynchronous	Data sent character by character with start/stop bits	No shared clock	RS-232, USB, Bluetooth

Example:

- **Synchronous:** Like a train (data bits) following a fixed schedule (clock).
- **Asynchronous:** Like sending letters one by one (each has start/stop mark).

◇ 9. Standard Communication Interfaces

Standard interfaces define how devices communicate with the computer.

Interface	Full Form	Description	Example Device
RS-232	Recommended Standard 232	Serial communication standard	Old modems, serial ports
USB	Universal Serial Bus	Fast serial connection, plug-and-play	Pen drives, cameras
SCSI	Small Computer System Interface	High-speed data transfer for storage	Hard disks, scanners
SATA	Serial Advanced Technology Attachment	Interface for connecting storage devices	HDD, SSD
PCI	Peripheral Component Interconnect	Connects internal devices	Sound card, network card

◇ 1. Programmed I/O vs Interrupt Initiated I/O vs DMA

Feature	Programmed I/O	Interrupt Initiated I/O	Direct Memory Access (DMA)
CPU Involvement	CPU actively checks device	CPU interrupted only when device ready	CPU not involved in data transfer
Speed	Slow	Faster than Programmed I/O	Fastest
CPU Utilization	Wasted (busy waiting)	Efficient (CPU free till interrupt)	Very efficient (CPU totally free)
Control	CPU controls transfer	Interrupt controller controls	DMA controller controls
Best For	Simple, small data transfers	Moderate data transfers	Large data transfers
Example	Keyboard input	Printer interrupt	Disk to memory transfer

◇ 2. Synchronous vs Asynchronous Communication

Feature	Synchronous	Asynchronous
Clock Signal	Shared clock used	No shared clock
Data Transfer	Continuous stream	Character by character
Speed	Faster	Slower
Synchronization	Sender & receiver synchronized	Start/Stop bits used for synchronization
Overhead	Less (no extra bits)	More (due to start/stop bits)
Example	SPI, Ethernet	USB, RS-232, Bluetooth

◇ 3. Hardware Interrupt vs Software Interrupt

Feature	Hardware Interrupt	Software Interrupt
Source	Generated by hardware device	Generated by instruction/program
Trigger	External signal	Software instruction (like TRAP)
Examples	Keyboard press, Disk ready	System call, Divide by zero
Control	Managed by interrupt controller	Managed by operating system
Priority Handling	Hardware-defined	Software-defined

◇ 4. Interrupt vs Exception

Feature	Interrupt	Exception
Source	External to CPU	Internal to CPU
Cause	Hardware device or software request	Errors or special conditions during execution
Examples	Keyboard input, Printer ready	Divide by zero, Page fault
Frequency	Asynchronous (any time)	Synchronous (during instruction execution)
Purpose	To handle I/O events	To handle errors or traps

◇ 5. Parallel Port vs Serial Port

Feature	Parallel Port	Serial Port
Data Transmission	Multiple bits at once	One bit at a time
Speed	Faster for short distances	Better for long distances
Number of Wires	Many	Few
Cost	More expensive	Cheaper
Example	Printer (old systems)	USB, RS-232

◇ 6. I/O Channel vs I/O Processor

Feature	I/O Channel	I/O Processor (IOP)
Nature	Special hardware controller	Independent processor
Instruction Execution	Executes I/O channel commands	Executes I/O instructions
CPU Dependency	Works under CPU supervision	Works independently

Complexity	Less complex	More complex
Example Use	Disk and tape control	Mainframe I/O handling

◇ 7. Memory-Mapped I/O vs Isolated I/O

Feature	Memory-Mapped I/O	Isolated I/O (Port-Mapped I/O)
Address Space	Shares same address space as memory	Separate I/O address space
Instruction Set	Same instructions for memory and I/O	Special I/O instructions used
Address Lines	Common for memory and I/O	Separate lines
Ease of Programming	Easier (uniform access)	Slightly complex
Example	ARM, modern processors	Intel 8085, 8086

◇ 8. Input Device vs Output Device

Feature	Input Device	Output Device
Function	Send data to CPU	Receive data from CPU
Data Flow	From user → system	From system → user
Examples	Keyboard, Mouse, Scanner	Monitor, Printer, Speaker

Notes by -GKM